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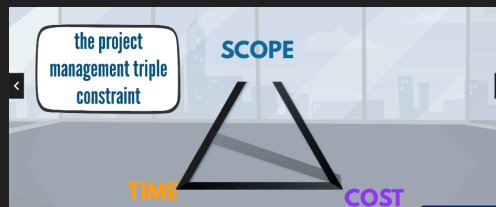
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Chapter 1

Welcome to the course! Introduction to projects and project management

1.1 What is a project?

- A project is:
 - A temporary endeavor undertaken to create a unique product, service or result. PMI definition.
 - It has a determined start and end date.
 - It is not a business process that is ongoing but rather have a specific expiring date.
 - It is time bound and has a goal.
 - The goal can be a result, service or product.
 - The output + the work employed to create it is called a scope.
- Elements of a project:
 - Time (start and end date) + goal (product, service or result) + cost.



- These three elements form what is known as a project management triangle where any change in one of the dimensions results in the changing of all the other dimensions, there is a tradeoff between these dimensions, when you want less time this might affect cost, etcetera.
- Ending definition:
 - A project is a temporary initiative that is agreed, planned, and executed to achieve a specific goal.
 - Projects are complex initiatives.

1.2 Why do companies execute projects?

- All companies have goals. Some goals are for daily work goals and other goals are strategic.

- To achieve the strategic goals a company cannot lean on the same old strategies but rather create something new.
- In the implementation of any strategic goal you need to achieve in a company, you must do it using a project. This is why companies execute projects.
- In order to achieve the project as an output you need to put in the following inputs:
 - Resources,
 - Financial Resources,
 - Organizational resources, and
 - Time.
- In highly innovative industries failing to prioritize getting the necessary resources in the right time might put the company at a significant disadvantage.
- Correct project management derives the following benefits:
 - Financial, operational, customer related, non-profit.
- Let's see the life cycle of a project:



1.3 What creates the demand for a project? Prioritizing and selecting a project.

- A project must relate to a business strategy. So what creates the demand for a project in the first place.
- The following:
 - Market need, to keep up or have advantage over the competition.
 - Business need.
 - Technological advancement. i.e. for example: automation.
 - Customer request. i.e. satisfying client's needs.
 - Legal requirements. For example: social media.
 - Social needs. For example: building a bridge.
 - Ecological impact considerations. For example: improving cars
- Once a specific goal arises, the PROJECT will be the instrument to achieve that goal.

1.3.1 Priritizing and project selection

- The needs most likely will never arrive one by one, so how do you choose which ones to do and in what order?
- This is when prioritizing comes in.
- The process is the following:
 - First the project owner will propose the project via a project proposal to management to make it compete for the resources of the organization.
 - Management will decide which project will be discarded, prioritized, or postponed for a later date.
 - Prioritizing urgency: for example if some law requires some immediate project or else they won't be able to operate, management will prioritize it because it is urgent.
 - This is known as project selection.
- The project selection process selects the order in which each project will be done, and will schedule the order in dates. Which projects will be concurrent etcetera. Management will make what is called a project portfolio, this is a document that contains the details on which projects will be done and in what order, and which project will be done at the same time (concurrently).
- Project Portfolio Management: the process of creating and managing the project portfolio in such a way that it uses the organization's resources in an optimal way.
- Projects are investments and need to be done in as efficient a manner as possible. Projects consume a lot of resources, finances, time, effort, attention, care, etcetera. If they are not aligned and optimized it may result in the organization's bankruptcy or at best to disappointed stakeholders.

1.4 What is a project manager? What is their commitment to a project?

- The project manager is the one responsible for the project, the PM is the CEO of the project.
- The PM person needs to work between these constraints (budget, time, performance) in order to meet the goal and add value to the business.



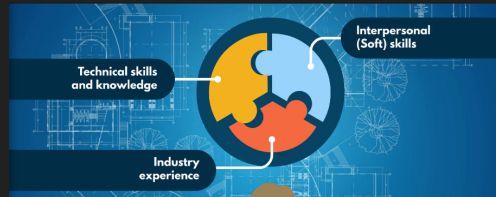
- This is a big responsibility. The PM must therefore choose realistic projects, taking into account all costs, all time constraints. Otherwise they will not be able to deliver and will be in problems.
- It is a project manager's duty to do the following to a project:

- Assess, Discuss, Negotiate, Adjust: demonstrate why the project will be ready in the proposed date, why it will cost the proposed amount. Otherwise refuse to take a project on.
- What does it mean to be accountable?
 - The project manager has the ultimate responsibility, they can't blame the failure of a project on any one else. 'The buck stops with them'.
 - Project manager must commit and deliver.
- The PM responsibility is not only responsible for themselves but also for the actions of their team members. A PM must be able to have full control, and take immediate action upon a problem arising.

1.5 What are the skills and the knowledge a PM must have?

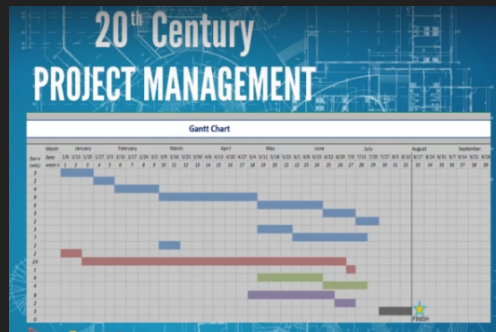
- The PM is the face of the project.
- They are the point of contact of the team.
- Must mediate between team members.
- Successfully relate with team members with all levels of seniority.
- They are accountable to deliver the expectations.
- They must therefore have:
 - Skills, knowledge, attitude, and experience.
- The three major qualities of PM:
 1. Technical skills and knowledge.
 - PM triple constraint.
 - Theoretical framework.
 - PM tools and documents.
 - Personal organization (ability to organize your own tasks).
 - Task management (ability to fulfill tasks, prioritize and manage tasks).
 - Critical thinking (tell which information is useful and which is not).
 - Efficient use of software tools for PM.
 2. Interpersonal soft skills:
 - Each project is unique and must adapt to a new set of persons each time.
 - They need communication skills (90% of their time will be in communications).
 - Team management, being able to assess performance and output of the team, the PM needs to be able to articulate the project's needs and requirements, identify impediments and keep the team motivated.
 - Leadership, ability to influence people in the desired direction, capacity to mediate.
 - Social skills, ability to work with a lot of people.
 - Advice: never talk exclusively about the project, talk also about personal lives and hobbies, otherwise team members will associate you only to office work and you will not be able to influence and motivate them personally. Also be humorous, it relaxes people.
 3. Industry experience:
 - Does the PM need to be an expert in the industry in which their project are taking place? it depends, knowledge always helps.

- Being an expert in a field does not imply success in managing the project related to that field. It helps but in order to manage project effectively you need first and foremost... project management skills. Industry expertise is a bonus most of the time.



1.6 Projects: a history lesson

- Project management has been going on for thousands of years.
- An example includes the pyramids of ancient Egypt. This construction was a massive endeavor, involving materials and a workforce. The success of this most likely depended on good project management.
- Another example is when Columbus sailed to America, he did it with a budget of three ships and some ship crew and his proposal was rejected multiple times by various kings. Eventually they found a sponsor of this project, the king and queen of Spain.
- Another example, 19th century and the industrial revolution, in which project management became more and more popular since more projects were going on.
- Another example, 20th century project management, the Gantt chart was created by Henry Gantt, it was used in WW1 to track and manage navy ship creation.

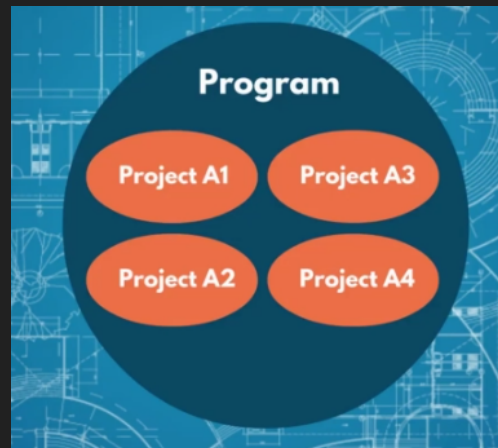


- In the late 1950s the *critical path method* this is a methodology that allows to calculate the fastest way to completing a project, finding links and connections between the activities involved.
- In the 1960s the Project management profession was established. The international project management association and project management institute established standards, best practices, tools and guidelines.
- The 21st century project management scheme is exceptional because the projects are incredibly complex and have many many variables and factors, as well as risks that must be all individually considered. A strong PM is a very important asset in all organizations.

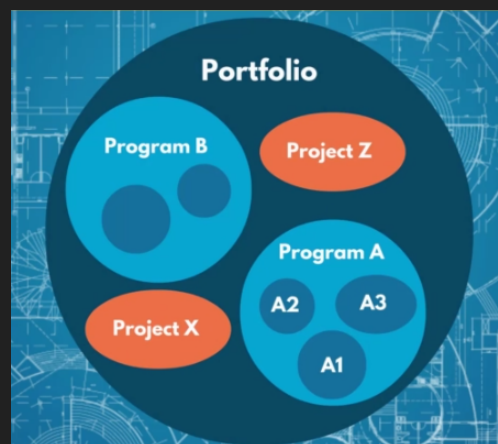
1.7 Project management terminology

- Project management office (PMO): department responsible for managing, coordinating and consulting project related work. The PMO will vary in size and structure from company to company.

- Consulting companies will have very large PMO unit. While some others might not even have a PMO unit.
- Project team: experts responsible for the execution of the work. Anyone directly working on the project.
 - Example: developers, supervisors, designers, engineers, etc.
- Project stakeholders: all individuals and/or organizations who participate in a project, are influenced by the project work and results. Stakeholders can influence the project work even if they are not involved.
 - Example: new underground line is being built near your house. You are a stakeholder even if you are not involved in the actual construction.
- Program management: the coordinated management of multiple projects, which have similarities - similar goals, similar resources, etc. This typically involves multiple PMs working with each other in multiple projects.



- Portfolio management: The coordinated management of multiple programs and projects.



- For example in a pharmaceutical company there might be several drugs being developed at the same time, this will be handled as different projects and inside a portfolio.

Chapter 2

The project phases

2.1 How does a project start and evolve? What a project's main stages?

- It has 5 phases:
 1. Initiation: the foundation of a project, a justification of why it is needed, this will include a suggested budget and time frame.
 2. Planning: create a plan in which the PM needs to estimate the work, risks, resources. Find out the activities, scheduling and timing, budget, project roles. All of the research done in this phase will be included in a document named project management plan. Planning = strategy for winning.
 3. Execution: the actual activities start, all the members of the team are assigned work.
 4. Monitoring and control: this does not start after execution but rather is done at the same time as the execution phase and answers the following question:
 - Progress done? how well are we doing with respect to the time frame? the budget used will be enough? any delays or gaps?
 5. Closure phase: the final tasks, handing the project to the person who will use it, the person receiving it must sign off and confirm that it is complying with the terms. Then the PMs, stakeholders and team sit down and share the lessons learned in the just finished project, feedback is collected and rest until the next project begins.

Chapter 3

The initiation phase

3.1 What is the initiation phase about? Define project goals

- Starts with an idea or need, the person who comes up or identify the need is the project sponsor.
- The responsibility of this person are the following:
 - Feasibility?
 - Do we have the resources to complete it?
 - Will the project deliver the expected benefits?
 - Appointing a project manager.

3.1.1 Goal

- Defining a goal, remember the definition of a project, that of a temporary initiative that is agreed, planned, and executed to achieve a specific goal.
- The goal must be worthy of being granted a project.

3.2 What is involved in a business case?

- A business case is a document intended to convince management to invest company resources in this project and not in somebody else's.
- It must outline the benefits and the cost (including taxes, inflation, etcetera), the purpose of the project. This is a document that the business leader and project sponsor prepares.
- A scope is defined, a scope includes all activities and questions that belong to the project. For example in building a showroom project you might ask whether or not taking pictures of the not yet completed showroom is part of the project, this is a gray area.
- A gray area is questions and situation which arise during the implementation of a project with respect to what is actually part of a project.
- The project team includes production, IT, R&D,
- A project manager must spot and say in advance and in the initiation phase whether or not something is realistic. If the PM finds a project not feasible the PM must decline to participate in the project.

3.3 Who performs the feasibility study and what does it involve?

- A business case details all the finances with respect to the project.
- A scope outlines what your project will involve.
- A feasibility study analyses the goals, scope and the resources to see if they are feasible.

3.3.1 Feasibility study

- The project sponsor and PM should be efficient and analyze this quickly to check generally if the budget is feasible, if the organization possesses the needed expertise, do partner companies need to be involved, etcetera.

3.4 What goes into risk assessment? What are the expectations?

- Assuming the project is feasible, there are always black swans in the world and bad luck can strike. What are the impacts and risk of this project? This is answered in the risk assessment document.
- A risk assessment also checks what can go wrong in order to plan in advance what can we do to mitigate those risks.
- This document is also used to outline the expectations, outline who is accountable.
- We must spend time with stakeholders in order to manage expectations and listen for potential gaps between what is actually being proposed and how the project is going to look once terminated.

3.5 How to create a project charter?

- Assuming the project is still viable despite the risk assessment and feasibility.
- A project charter is a high level document that is quick and easy to understand. An example:

365 Careers				Project Charter			
Business Need / Project Goal							
Project Scope							
In Scope				Out of Scope			
Key Risks							
Key Project Milestones							
				1			
				2			
				3			

- You can also include the business cases in this document. The important thing is to make this document as easy to read as we can, any one can read it, find what they need and move on. If what they need is more details, the charter will show them where to find the details they need.

3.5.1 Recap

- Initiation phase is the foundation for the entire project.
- The PM can be appointed at any time during the initiation phase so the steps in the initiation phase do not need to be executed in chronological order.

3.6 Case study 1

Outlines the story of an insecure BA that transformed herself into a PM and realized the following:

- No expertise in anything but PM is required to be a PM in a project.
- A PM needs to adapt to new environments quickly.
- A PM is always learning.

Chapter 4

The planning phase

4.1 What is planning? Why is it important? What happens if it is not done correctly?

- Planning: process of analyzing, evaluate, decide and organize activities in advance.
- Three fundamental steps: define the goal, evaluate options, choose and confirm the best option to achieve the goal.
- Planning is on 2 levels: strategic and tactical.
- Failing to plan properly can end in terrible results.

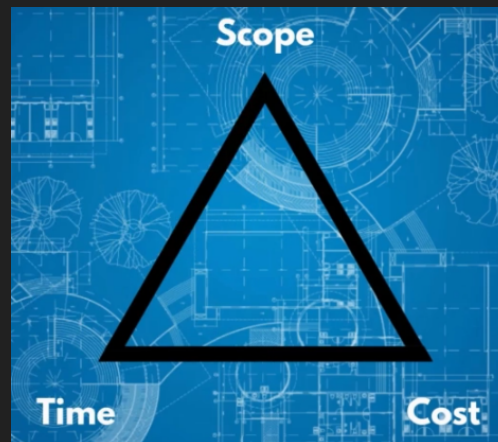
4.2 Why is planning so critical in PM?

- Planning is the fundamental work of the project manager.
- It is when the PM demonstrates their expertise.
- You can divide planning in 2:
 - Things the PM does not know.
 - Things the PM does know and can't control. Here risk assessment, estimations and forecasts need to be covered, this is done using planning.
- Good planning gives great results: optimize work, ensures same expectations, avoids costly errors. The better planning the more chances of success the project has.
- People also work much better with direction, when they understand what to do, how to do it and when they must do it.
- Planning is your strategy to succeed. Despite this planning is often criminally underestimated. People think that the sooner they execute (without planning) they better but in reality the more effort is expended in the planning phase the less effort is needed in during the execution phase. This is the 80/20 rule. More planning means less replanning and delays during execution.

4.3 What is the cost of change in projects?

- Each change to the original project outlined in the project charter impacts cost and time, often both. The additional budget, time, resources, and coordination efforts are the "cost".

- Remember the triangle of constraints:



- When the change is made also impacts, since it is cheaper if the change is thought out during the initial phase of the project than if the project is near the end.
- The more time passes during execution of the project the higher cost of change it has:



- A PM must take into account potential changes the project might need in advance and that is why they need to be master planners.

4.4 What to do before you start?

- There are some considerations the PM needs to take into account during planning:
 1. The PM must assess their own knowledge of the required work. (Is it enough to plan?)
 2. The PM must know all the project stakeholders.
 3. The PM must recall lessons learned. (Has this project been done before? If so what lessons can we get from the previous projects?)
 4. What about knowledge gaps? (a good project manager is not someone who knows everything, but someone who can identify their own shortcomings and find the right people to compensate for them.)
- A PM must determine how much time they need to put all the pieces of a project together. Also decide with who they will keep in contact with. And also reaffirm the expectations to stakeholders.
- In meetings with stakeholders they must prepare with an agenda, a goal, and a comprehensive list of topics to cover.
- PM need to plan for everything.

4.5 Project management insights

- Changes are frequent during planning.
 - New information can arise at any time.
- Planning simple tasks:
 - A simple task must be planned too.
 - A meeting needs to be planned out by determining the participants, day time, room for the meeting, flipchart, and sending out the invitations.
- Tasks with unplanned time durations:
 - Some tasks you cannot estimate, especially if they depend on other people to come to fruition. These tasks can become problematic and can delay the project if not managed correctly.
 - Example of hiring an engineer, you write to HR, after a day they respond they need more details, you reach out to IT in order to get the details, 2 days pass and you have the details, but the details need to be verified by the head of IT that is in vacations and thus what could have easily been planned to take 2 days become 8+ days.

4.6 Scope planning

- At this stage the PM should have information enough to start planning. The planning has its own structure.
- The first thing a PM needs to do is to define the project scope.

4.6.1 Setting up a scope

- Establish the questions that need to be answered.
- In the lamborarri showroom example we need to define how many floors will it have, who will be in charge of what, etcetera.
- Remember scope means what exactly does the project team need to do.



- Once the goal has been established we can then define deliverables.
 - A deliverable must have all the requirements clearly established.
 - An example of deliverables is the following:



- For example let's say that in the land in which we plan to build the show room there is already a building, we need to factor this into our scope since we cannot do anything related to the actual project until we demolish the building.
- Product scope: the tasks that need to be done to produce the product.
- Project scope: all that you need to do in order to achieve the project goal. This includes the product scope.
- A scope needs to be described in detail, a scope definition needs to include what is not in the scope also.
- Take into account that we are not looking for cent precision in planning, if you forgot some trivial thing then that can be added later on with no major incident but if you forgot something big such as a million dollars worth of details then that will be a problem.
- In the showroom project this needs to be agreed between the construction company and the city.

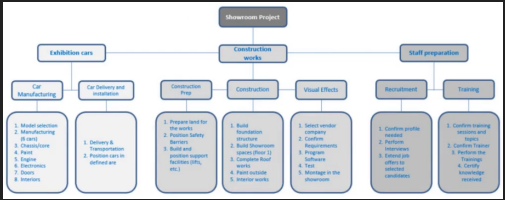
4.6.2 Scope steps

- Assessing information on the project scope.
- Gathering details on project requirement and expectation.
- Using expertise to cover grey areas.
- Documenting everything. (If something arises that need to be checked then it must be available.)

4.7 Scope part 2

- Recap:
 - We analyze the available information.
 - We gather detailed requirements and expectations.
 - We document the scope.
- Now for a low level look at scope planning:
 - The PM analyses high-level information gained in the initiation stage from the project charter, discussions with the sponsor and clients, as well as stakeholders.
 - Then gather detailed requirements and expectations, such as the sponsors and managers, the sponsor and senior manager are a reliable source of information on scope if the PM asks the right questions.
 - The more information the PM gathers the better.
 - Then with all this information the PM must document all this in a document called a scope statement document. This document is looked at as a sort of contract, between the PM and stakeholders, it outlines what the PM has committed and has not committed to do.
- The scope statement document is not user friendly.

- This is why the scope statement needs to be reformed and split into deliverables and then each deliverable must be assigned to a specific person.

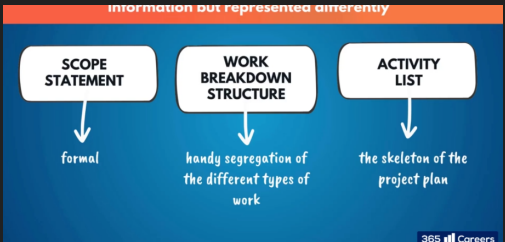


- For an even more user-friendly experience the PM creates an activity list. Which details the task, person, activity, etcetera. This to make it absolutely clear what will be done, by who and by which date. This document can be referred to by anyone.

the PM creates an activity list

Activity	Person	Activity
1. Assemble and position utility services		
2. Clear building site (demolition/reconstruction, removal of objects)		
3. Build and position support structure (PPS, etc.)		
4. Formwork foundations (digging)		
5. Construct main load-bearing structure		
6. Install utility systems		
7. Plumbing		
8. Sewerage		
9. Electrical wiring		
10. Communication fibres		
11. Thermal insulation		
12. Build underground wall floor (concrete and steel construction)		
13. Build underground walls (concrete)		
14. Build ground level floor		
15. Build entrance space (Phase 1)		
16. Construct main load-bearing structure		
17. Build bearing walls		
18. Wall framing in house construction		
19. Build interior spaces (rooms and patios)		
20. Complete roof works		
21. Build top floor / roof (concrete and steel construction)		
22. Install water-resistant barrier		
23. Install cladding for flat roof		
24. Main construction completed		
25. Install doors and windows		

- Note that there are three documents involved here that contain the same information in different formats:



Chapter 5

The planning phase - timelines and schedules

5.1 How to estimate when planning?

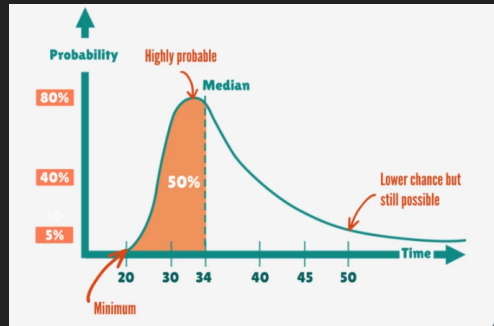
- Remember the three documents that have the same information but represented differently:
 - The scope statement: formal definition of everything we need to do.
 - Work breakdown structure: user friendly description of the work to be completed.
 - Activity list: skeleton of the project.
- Now we will learn to estimate the activities.
- To build a project timeline we need to estimate durations of each task, add buffers, and sequence the tasks.

5.1.1 Estimating the durations

- The PM needs to estimate in time each activity, as a PM you must do all the necessary research to estimate correctly, you can ask the team to estimate but it is your responsibility the time you choose. Remember you are the accountable one.
- The PM should not take the team's opinion as the ultimate truth, the PM must do their own research and estimate conservatively.
- The PM can collect statistics on team members to estimate the time correctly and accurately. The idea is to base your estimation as much on data and minimally on opinion.

5.2 The planning fallacy, optimism bias, illusion of control

- Take into account that we are all susceptible to the planning fallacy.
- The planning fallacy is the idea or notion that we plan and estimate based on optimism and we all tend to underestimate the amount of time (and cost) it will take to complete a task. And predictions are often wrong because of this.



- The planning fallacy was discovered by Daniel Kahneman and Amos Tversky in 1979.
- One of the key parts of the planning fallacy is the illusion of control, the idea that we control events.
- People are generally bad at planning.
- As a PM you must be aware of these human biases.
- Estimation is a complex activity.

5.2.1 The three point estimate

- When a PM is stuck and cannot estimate correctly there is a formula you can use.
- Identify these three components:
 1. O as the optimistic time estimate.
 2. M as the most likely or normal time estimate.
 3. P the pessimistic time estimate.

$$T = \frac{(O + 4M + P)}{6}$$

- Even this has a 50% of finishing before and 50% chance of finishing after. This is where buffers come in.

5.3 How much to buffer?

- First we need to account Murphy's law: "anything that can go wrong will go wrong."
- We don't control all so the adaptation to this law is to structure the project in such a way that if the worse does happen this will cause the least amount of damage. Using plan B (B stands for buffer).
- When estimating a time line the PM needs to put additional time buffers that cover possible delays:

Task A- estimated time needed	Buffer	
Start Date	Expected Completion	Official Deadline

- Buffer may be added to individual tasks, groups of tasks, activities or a projects.
- Choosing the buffer is tricky but the following guidelines are helpful.

5.3.1 Buffer estimation guidelines

1. The higher the level of uncertainty the longer the buffer.
 - Example.



- For example if you are workign with people you have never worked before, you must take into account the possible problems that might reveal, and the buffer must reflect and cover that.
 - The key word is accountability here.
2. If there is a task that will damage the project if delayed, then a longer buffer would be sensible.
 - The criticality of some tasks will be higher because if delayed they delay all the others that follow them.
 - One way to determine this is using the critical path chart (which we will see later).
 3. The owner of a task is a factor:
 - The experience factor plays a role. Someone with experience will be able to work better than someone without experience. Simply because there is a higher degree of uncertainty with the inexperienced.
 4. Avoid having excessive buffers and distribute buffers intelligently:
 - The buffers should be distributed across the project's most uncertain activities, putting all the buffers at the end for example is a bad practice and odds are you will need to decrease it because the stake holders will think its idle time.
 - Keep in mind that presenting the buffers as buffers may lower productivity because the team thinks that once they have finished before the buffer it's ok to be idle until the next activity is required to start. A better way is to present it as a validation period or safety check.

5.4 Identifying dependencies

- By now the project should have a work breakdown structure, an activity list, and an estimated time per task along with buffers. Now we must check what activity is dependent on another. So that we can parallelize tasks that don't depend on each other.
- There are types of dependence outlined belowed.

5.4.1 Types of dependencies

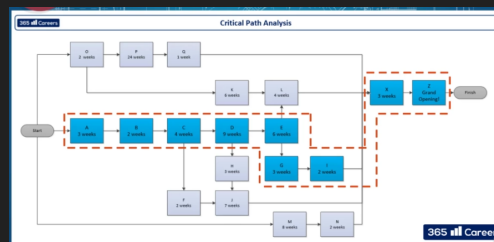
1. Logical dependency: for example you cannot install windows without having walls.
2. Resource constraints: sometimes all though some tasks can be parallelized there just isnt enough resources to perform them at the same time. If you onnly have 1 business analysts and you need a business analysis done, you cannot expect the BA to perform this simultaneously.
3. External constraints: activities affected by external factors such as weather. For example painting cannot be done in a rainy day.
4. Soft dependencies: dependencies that change during program execution

5.4.2 Other types of dependencies

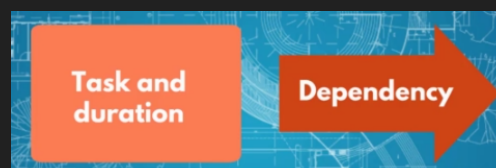
1. Finish to start: second task cannot start until the first task is finished (waterfall).
2. Finish to finish: second task cannot finish until the first task has been finished.
3. Start to start: the second task cannot start until the first starts.
4. Start to finish: the second task has to start before the first can finish. An example is a shift of guards, the guard who is starting his shift must start before the guard who is leaving finishes his shift, this allows no windows for thieves to break in.

5.5 Identifying the critical path

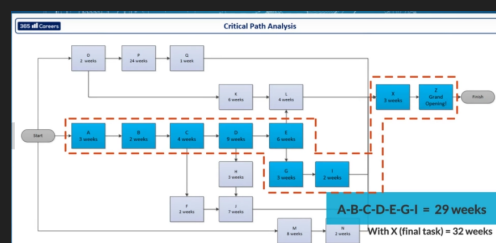
- Given the completed list of activities with their buffers.
- So now you must identify the dependencies across the activities.
- Next you identify the longest chain of dependant activities as outlined in the following image:



- To start this use a network graph that identifies the dependencies across activities and then turn that into a critical path chart. Each box in this diagram will be a task with its duration, and each dependency will be a line connecting the dependant activity.

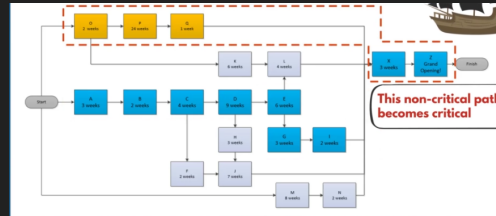


- Given a work breakdown and the activity list.
- Now the longest chain of activities is the critical path, since any delay here will postpone the whole project.



- With the non-critical paths you have time, a delay most likely will not halt the whole project. This time that we are calling 'extra' is called floating time.

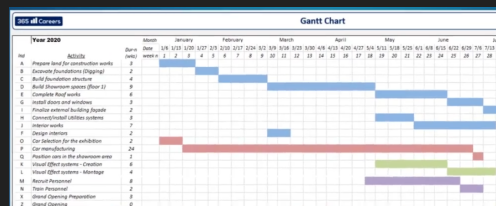
- Another very important thing to consider is that the critical path can change when there is a delay in any non critical path. If the delay is big enough and overlaps with other activities, this means that resources may become cosntrained. For example in the below image we see that originally the middle path was the critical path but since there was a delay in a non critical task now due to that delay the new critical path will be the delayed task. This is partially why buffers are so important.



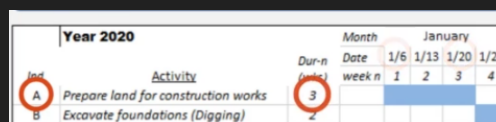
- Now we may as well call the project schedule done, however there is one thing missing. The gantt chart.

5.6 Using the Gantt chart to plan the project work streams

- The critical path chart seems a little messy and we cannot visually tell when each activity starts and when it ends.
- To solve this we use the Gantt chart. It looks sort of like this:



- The idea is to list in the rows the activities pending and in the columns the time. Then the tasks we will parallelize will have their own space to be placed and it will be easy to see which task needs to start at end at which date.

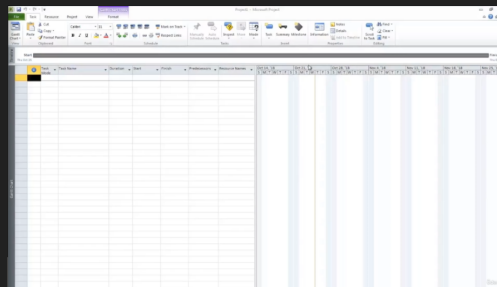


- We can now visualize the duration and the start at end date clearly. Strategizing and optimizing is easier now.

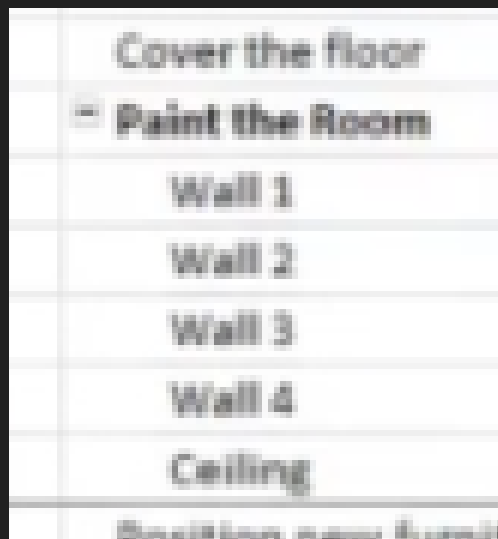
5.7 Building the project schedule (+ MS Project tutorial)

- All of the documents seen so far (the business case, project charter, work breakdown, activity list, gantt chart) is really for this last document. Which is the master document. The project schedule document.
- We will be using microsoft project.

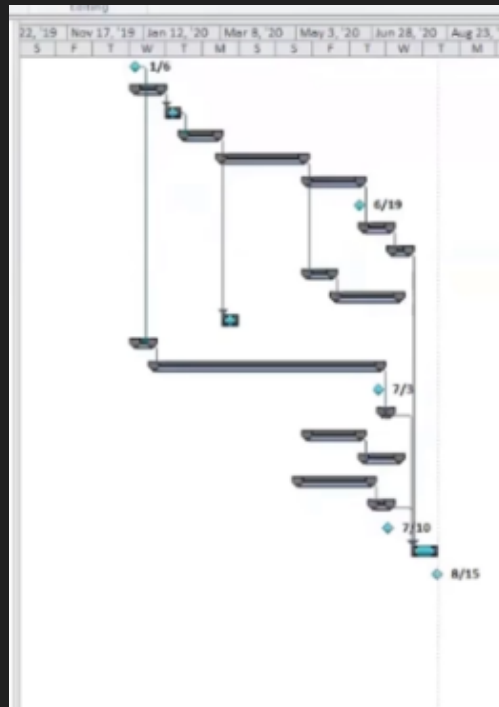
- It looks like this:



- Notice the components, the left pane is for the tasks and their details and the right pane is a gantt chart that is automatically populated when you put in the dates and details.
- In the field predecessors is where you indicate dependencies.
- Resource names is for names or roles of the person in charge of the task.
- To create subtasks you just need to select the tasks and indent them, this will make them subtasks to the one above.



- See the automatic gantt chart of all the non-sub-tasks:



- To see the subtasks we just expand the subtasks in the left pane and this will automatically reflect the subtasks in the right pane.

5.8 How to build a milestone table and its uses

- As you can imagine the ms project template can be thousands of rows and columns in extension so when any non-project manager may find this overwhelming so again we can summarize this to give the stake holders updates on the progress of the project.
- The milestone table document is a table that is meant to be a high level summary of the progress of the project. It includes actions, deliverables and project milestones.

365 Coreers Milestone Table			
Activity / Milestone	Start Date	End Date	Status
Project Kick Off	6-Jan	6-Jan	Not started
Prepare land for construction works	6-Jan	24-Jan	Not started
Excavate foundations (Digging)	27-Jan	7-Feb	Not started
Build foundation structure	10-Feb	6-Mar	Not started
Build Showroom spaces (floor 1)	9-Mar	8-May	Not started
Complete Roof works	11-May	19-Jun	Not started
Main construction Completed	19-Jun	19-Jun	Not started
Install doors and windows	22-Jun	10-Jul	Not started
Finalize external building façade	13-Jul	24-Jul	Not started
Connect/install Utilities systems	11-May	29-May	Not started
Interior works	1-Jun	17-Jul	Not started
Design Interiors	9-Mar	20-Mar	Not started
Car Selection for the exhibition	6-Jan	17-Jan	Not started
Car manufacturing	20-Jan	3-Jul	Not started
Car manufacturing Completed (6 vehicles)	3-Jul	3-Jul	Not started
Position cars in the showroom area	6-Jul	10-Jul	Not started
Visual Effect systems - Creation	23-Mar	1-May	Not started
Visual Effect systems - Montage	22-Jun	17-Jul	Not started
Recruit Personnel	4-May	26-Jun	Not started
Train Personnel	29-Jun	10-Jul	Not started
Personnel preparation Completed	10-Jul	10-Jul	Not started
Grand Opening Preparation	27-Jul	14-Aug	Not started
Grand Opening	15-Aug	15-Aug	Not started

- So many documents, how does this one fit in? Let's recap.
 - A PM starts by creating a work breakdown structure.
 - Then an activity list, from this list we identify the dependencies.
 - Then the gantt chart reflects the execution of each activity and can choose which tasks aren't dependant on each other so that they can be parallelized.
 - Then the PM will create the high-level milestone table and the low-level project plan (schedule).

Chapter 6

The planning phase - cost

6.1 The process of budgeting

- Planning is not a simple job when it comes to time, similarly it is complicated when it comes to cost.
- Budgeting seeks to answer questions such as:
 - What financial resources do they need to create the deliverables?
 - Are there going to be additional expenses?
 - Will they need the money?
 - Where will it come from?
 - What is the total budget?
 - Who will control the costs? how?
- The most convenient way of answering these questions is the task of the project budget list.
- Building the project budget list follows a similar structure to building a project schedule. Here are the steps:
 1. Identify the activities that cost money from your work breakdown structure or activity list.
 2. Estimate the cost of each work package, task and deliverable.
 3. Add buffers and account for uncertainties.
 4. You should be ready to time the expenses that can be done using a gantt chart.
 5. Put it all together into a project budget.
- Remember the optimism bias, just like in planning time humans often underestimate the time taken, humans tend to underestimate cost too.
- The PM needs to be accurate and avoid the optimism bias.

6.2 The process of budgeting (continued)

- The first step of cost planning is identifying all the cost generating activities.
- You need to get your work breakdown structure and activity list. Using the activity list check for potential costs that will not be covered by the company, these costs will be most likely covered by someone else, such as another company that we hired to do certain jobs such as paint walls and such. Who will buy the paint? us or the painting company? This all needs to be agreed upon before anything starts and it must be mentioned and agreed upon in a contract.

- You need the project sponsor to advise how to factor the cost. Communicating with the appropriate people is crucial.
- You need to be in contact with HR for hiring and answering questions such as the following:
 - How many staff members are needed?
 - What jobs they will have?
 - What costs will be incurred for recruitment?
 - Who will train the new hires? You or another company?
 - Will they be interviewed on site or in an external center?
- If you have built a detailed and extensive activity list them identifying which ones incur costs should be a lot easier. Budget planning may make you aware of tasks that you did not factor into the project schedule.
- Now that we have a list of all the costs, it is time to estimate what those costs will be. Some tasks will be straight forward to estimate but others will not, some will require negotiation with other companies, others will require contacting many people and negotiating.
- Using information from past projects is a very useful tactic.
- Take into account the optimism bias and adding buffers.
- Track all the costs and spending in a project budget sheet.

Budget (Overall Project/Program)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Total Budget (over activity)
Project Overall Budget												
Prepare land for construction works	\$ 4,000											\$ 4,000
Acquire foundations (ongoing activity)	\$ 10,000											\$ 10,000
Build foundation structure		\$ 200,000	\$ 200,000									\$ 400,000
Build drainage system (over 10)		\$ 400,000	\$ 400,000									\$ 800,000
Complete roof works				\$ 120,000	\$ 120,000							\$ 240,000
Design interior		\$ 10,000										\$ 10,000
Install doors and windows				\$ 70,000	\$ 70,000							\$ 140,000
General build (interior systems)				\$ 20,000	\$ 20,000							\$ 40,000
Finish external building facade						\$ 30,000						\$ 30,000
Interior works						\$ 10,000	\$ 10,000					\$ 20,000
Visual Effect Systems - Creation				\$ 20,000								\$ 20,000
Visual Effect Systems - Montage					\$ 10,000							\$ 10,000
Manual Personnel				\$ 10,000	\$ 10,000							\$ 20,000
Non Personnel					\$ 5,000	\$ 5,000						\$ 10,000
Car selection for the exhibition	\$ 2,000											\$ 2,000
Car transportation		\$ 40,000	\$ 40,000	\$ 40,000	\$ 40,000							\$ 160,000
Position cars in the showroom area						\$ 5,000						\$ 5,000
Visual Opening Presentation						\$ 20,000						\$ 20,000
Event Planning							\$ 20,000					\$ 20,000
Total Budget (over Monthly)	\$ 16,000	\$ 210,000	\$ 260,000	\$ 180,000	\$ 205,000	\$ 365,000	\$ 365,000	\$ 30,000	\$ 30,000	\$ 30,000	\$ 30,000	\$ 2,400,000

6.3 Procurement contract types

- It is often the case that the company will not have the resources they need for a project, so a PM must look elsewhere for the resources.
- The PM needs to be aware of the timelines, due dates and the quality levels of all resources expended and estimated.
- There are three types of contracts a PM can use for this.

6.3.1 3 contracts

- Fixed price: The vendor commits to doing the work for a set of amount within a certain timeframe.
 - Pro: any additional spending will be taken on by the vendor, no risk for us.
 - Con: Before signing the contract, they will most likely put a big buffer in the initial price. Or if costs do get too high they might decrease the scope or the quality in order to compensate.
 - This beneficial to use when the scope is clear to all parties.
- Cost plus contract:
 - The buyer agrees to pay any cost incurred by the vendor performing the work.
 - This can be a variable fee, a fixed additional fee or a mix.

- The risks here is that the buyer incurs all of the extra costs, the vendor can also keep working longer than needed or adding extra items to the 'to-do list'.
- This contract makes sense if the scope is not easy to define, but controls must be put in place in order to avoid over spending in non-essential things for the project.
- Time and materials contract:
 - It's a mix of the fixed price and cost plus.
 - The vendor charges the buyer a hourly or daily rate.
 - This is a good option when the scope is not clear and the work is more labor based than material based.
 - Similar risk to the cost plus contract, thus set controls to avoid the risks (overspending in non-essential things).

6.3.2 Conclusion

- There are many variations that can be tailor-made.
- You can even include inflation if the project spans a long time.
- You can add or decrease incentives as needed.
- The PM must have a good understanding of all risks and benefits that each agreement can have.

6.4 Procuring non-financial resources

- It's not all about the money in PM.
- Non-financial resources, which is to say humans, are also needed.
- Now for another analysis.

6.4.1 Analysing non-financial resources

It has four parts:

1. Define the type of resource:
 - Take your work breakdown structure and see what expertise you need, if the expertise can be found within the organization or extra hiring will be required.
 - Also allocate time to check management availability, if something needs to check something, do not assume they are always available, allocate some days for them to get to it. They are busy people.
2. Estimate how much time you will need.
 - Estimate the time each expert will need to complete the activities.
 - Take into account the optimism bias and adding buffers.
3. Validate the availability of the people you need.
 - (From next lecture: Planning HR)
 - You need to check when the experts can give you their time.
 - You can make a schedule with the person who needs to complete the activities and by when.

- Its important to keep track of employee schedule. And be in constant contact with the department heads, because sometimes they already have other activities and cannot give you any time, or refuse you access to them for any reason. If this is the case, you will need to work out a new plan with the project sponsor.
- Also very important: don't forget to check their personal availability such as vacations, holidays, etc.
- Creating a schedule looks like this:

The image shows a screenshot of a spreadsheet titled "Resource Availability Tracker". It has columns for tasks, responsible persons, and a calendar grid for tracking availability. The tasks listed include "Software Development / coding", "Software Development / testing", and "Testing of delivered software". The responsible persons are "John", "Alice", and "Mark". The calendar grid shows days of the week and months, with some cells marked as "On leave" or "On holiday".

- List the tasks in a column, then who is responsible for them, the duration. Then a calendar with all the marked vacations, holidays, and non-availability days.

4. Assign roles and responsibilities to them

- (From next lecture: RACI matrix - assigning roles)
- To assign roles and responsibilities we can use several tools, the first one is a simple list:

Role	Name	Responsibilities	Contact
Project Sponsor	Johnson	- Accountable for the project success - Support PM in resolving high-level issues - Owner of the Budget, Timeline and Business Case. To approve any changes.	J.Johnson@company.com
Project Manager	Philips	- Accountable for the project planning, execution and closure. - Daily management of project team and activities - Manage resources, risks, communications, stakeholders	P.Philips@company.com
Functional Team Lead – Software development	Matthews	- Provide status updates (progress, variances, potential issues, etc.) to Project Manager - Supports, guides and coordinates team members in completing their daily project tasks - Support other ad-hoc project tasks	M.Matthews@company.com
Team Member – Software Development	Henry	- Works directly to create the project deliverables (scope) - Uses subject matter expertise - Provides progress updates, communicates potential issues, seeks support - Suggests improvements - Support other ad-hoc tasks (e.g. testing, trouble shooting, etc.)	H.Henry@company.com

- It just has the role, name and responsibility along with the contact info.
- This is a people oriented document that is high level and simplified.

- The second tool is the RACI matrix:

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"RACI" Matrix

Activity / Deliverable	Project Sponsor Project Manager Marketing Director Marketing Analyst Production Director Chief engineer					
	PMO	Marketing Department		Production Department		
Project Management						
Appoint PM and support project	A		I		I	
Project Planning and Kick-Off	C	A/R	C		C	
Weekly Status Report	I	A/R	I	C	I	C
Work-Stream 1: Marketing						
Marketing Study		A	C	R		
Marketing Report with findings	I	A	C	R	I	
Work-Stream 2: Production						
Analyze future capacity needs		I			R	C
Build additional capacity / upgrades	I				A	R
Work-Stream 3: Supply-Chain						

R - Responsible

A - Accountable

C - Consulted

I - Informed

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- Each letter of RACI stands for something.
 - R: responsible, the person who is directly responsible for doing the work.
 - A: accountable, the person responsible for meeting the overall goal of the activity.
 - C: consulted, the person who provides information expertise or support for an activity.
 - I: informed: the person who needs to be informed on the status of an activity.
 - For most of the activities the PM will be accountable.
 - This is a lower level document compared to the simple list since more details are included and it is more formal. This is task oriented.
- The PM must choose which tool to use either one or the other or a mix of both.

6.5 Quality requirements

- Quality: the standard of something as measured against other things of a similar kind. The degree of excellence of something.
- In PM quality means: the quantitative characteristics of a product, either physical qualities, performance or service related.
- The PM needs to work with stakeholders to ensure that quality standards are being met on the deliverables.
- There is a five step process:
 1. Define the quality standards and requirements (could be according to laws or regulations too).
 - The PM needs to ensure standards are met.
 - Depending on the project, we might need to outsource or work with other companies for the quality assurance aspect. This is not a straight forward path since there is negotiating and constant issues that need to be spotted and corrected.
 2. Setting the actual, and actionable, targets for the criteria.
 - Set goals like: producing product with less of 0.5% defects.
 - Setting a tolerance level and using statistical methods helps ensure the quality goals are being actually met.
 - When it comes to services we need to set similar goal measuring metrics such as in a call center: less than 5% calls lost per month, or over 98% of incidents solved.
 3. Plan how, when and who will measure quality.
 - The PM must appoint someone to measure, and also must have clear what metrics we will use (as indicated in the previous step).

- The PM must audit the quality assurance to check whether the quality assurance metrics are actually being checked and passed.
4. Finalizing the quality plan:
- The PM collects all the information and creates yet another document, this document must include:
 - * Quality standards, target result, sample size of audits, frequency of reports, owner.
5. Make adjustments to their project constraints:
- A quality plan can increase the scope, the cost or the time for project completion in a number of ways.
 - Just in case any quality checks are not passed, a PM must identify risks and put buffers for when this happens so that the product, service and project comply with standards, this might require additional time and money so we must take this planning also into account.